κChow-functions

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Abstract.

1. Introduction

1.1. Main results.

2. Framework

2.1. **Posets and Incidence Algebras.** Let $\mathcal{P}$ be a *finite partially ordered set* ranked by the function ρ. We denote by $\mathrm{I}(\mathcal{P})$ the incidence algebra of the poset $\mathcal{P}$ over the ring $\mathbb{Z}[x]$ with the algebra structure given by the convolution product. Some of the functions we consider in $\mathrm{I}(\mathcal{P})$ are δ; ζ; its inverse, the Moebius function, μ; the characteristic $\mathcal{P}$-kernel χ and its associated functions. Following [FMV24], some of the functions associated to χ are the left and the right Kazhdan–Lusztig–Stanley functions (g and f respectively); the Z function; the reduced characteristic $\mathcal{P}$-kernel $\overline{\chi}$; the characteristic Chow function H; and the left and the right augmented Chow functions (G and F respectively). All these functions live inside a special subalgebra $\mathrm{I}_\rho(\mathcal{P})$ which consists of all the functions $\mathrm{a} \in \mathrm{I}(\mathcal{P})$ that satisfy $\deg(\mathrm{a}_{st}(x)) \leq \rho_{st} := \rho(t) - \rho(s)$ for all $s \leq t$ in $\mathcal{P}$. This subalgebra $\mathrm{I}_\rho(\mathcal{P})$ is equipped with an involution defined by $\mathrm{a} \mapsto \mathrm{a}^{\mathrm{rev}}$ with $\mathrm{a}^{\mathrm{rev}}_{st} := x^{\rho_{st}}\mathrm{a}_{st}(x^{-1})$. We summarize some properties of these functions as follows:

(1) $\chi = \chi^{\mathrm{rev}} = \mu \cdot \zeta^{\mathrm{rev}}$,
(2) $g = \zeta^{\mathrm{rev}}$,
(3) $\mathrm{Z} = g^{\mathrm{rev}} f = g f^{\mathrm{rev}}$ satisfies $\mathrm{Z}^{\mathrm{rev}} = \mathrm{Z}$,
(4) $\overline{\chi}_{st}(x) = \begin{cases} \chi_{st}(x)/(x-1) & \text{if } s < t, \\ -1 & \text{if } s = t, \end{cases}$
(5) $\mathrm{H} := -(\overline{\chi})^{-1}$ satisfies $\mathrm{H}_{st}(x) = x^{\rho_{st}-1}\mathrm{H}_{st}(x^{-1})$,
(6) $F = \mathrm{H} \cdot f^{\mathrm{rev}}$ satisfies $\mathrm{F}^{\mathrm{rev}} = \mathrm{F}$,
(7) $G = g^{\mathrm{rev}} \cdot \mathrm{H}$ satisfies $\mathrm{G}^{\mathrm{rev}} = \mathrm{G}$.

2.2. **The Hecke Module.** Let $R := \mathbb{Z}[x, x^{-1}]$ be the ring of Laurent polynomials with integer coefficients. The ring contains the additive subgroups $\mathrm{Pal}(n)$ consisting of all Laurent polynomials p satisfying $p(x) = x^n p(x^{-1})$. We note that these subgroups satisfy $\mathrm{Pal}(n) \cdot \mathrm{Pal}(m) \subset \mathrm{Pal}(n+m)$ and, in particular, $\mathrm{Pal}(0)$ is a ring.

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References

[FMV24] Luis Ferroni, Jacob P. Matherne, and Lorenzo Vecchi, *Chow functions for partially ordered sets*, arXiv e-prints (2024), arXiv:2411.04070.

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